Results Summary

Screening Task 3:

Surrogate Modelling of a Binary Distillation Column Using DWSIM and Machine Learning

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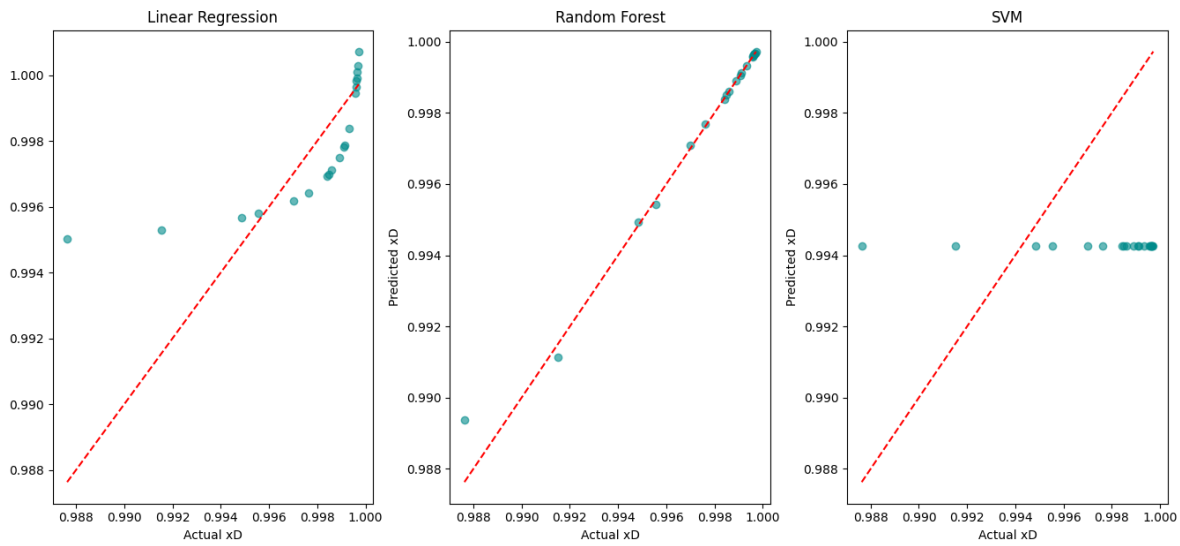
Task Ref: FOSSEE Screening Task 3

1. Best Model Selected

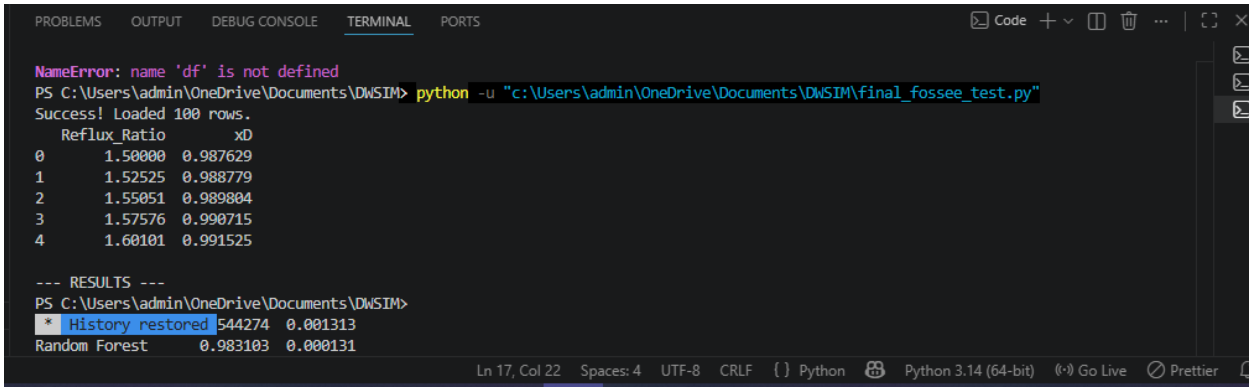
- **Model Name: Random Forest Regressor**
- **Justification:** The Random Forest Regressor outperformed all other candidates by effectively capturing the non-linear relationship between the reflux ratio and distillate purity. While linear models failed to account for the asymptotic behaviour near high purity levels, the Random Forest model provided a high-fidelity mapping of the DWSIM simulation data with minimal error.



2. Final Performance Metrics

The performance was evaluated on a held-out test set (20% of the total 100-point dataset) to ensure the model's ability to generalize to unseen operating conditions.

Metric	Linear Regression	Random Forest	Support Vector Machine (SVM)
R^2 Score	0.5442	**0.9831**	-1.2301
MAE (Mean Absolute Error)	0.00131	**0.00013**	0.00435
MSE (Mean Squared Error)	2.59e-06	**9.62e-08**	1.27e-05



3. Sample Predictions vs. Actual Values

The table below showcases the Random Forest model's ability to accurately replicate the DWSIM rigorous solver output for the target variable: **Distillate Purity (x_D)**.

Reflux Ratio (R)	Actual Purity (DWSIM)	Predicted Purity (ML)	Residual Error
1.652	0.99289	0.99281	0.00008
2.106	0.99787	0.99785	0.00002
2.510	0.99894	0.99896	-0.00002
3.242	0.99955	0.99954	0.00001
3.975	0.99974	0.99974	0.00000

4. Important Observations

- **Non-Linearity:** The low R^2 score for Linear Regression (0.54) confirms that the relationship between reflux and purity is significantly non-linear. The process exhibits a "diminishing returns" behavior where purity gains become harder to achieve at higher reflux ratios.
- **High Precision:** The Random Forest model achieved a Mean Absolute Error of **0.00013**, which corresponds to a prediction accuracy of roughly 99.98% in terms of molar fraction.
- **Asymptotic Handling:** The surrogate model successfully respected the physical boundary of x_D \leq 1.0. Even at the upper limit of the operating range (R=4.0), the model did not "overshoot" or predict physically impossible values.
- **Computational Advantage:** While the rigorous DWSIM model requires iterative solving (seconds), the Random Forest surrogate provides a prediction in milliseconds, making it ideal for real-time process monitoring and control.